

Notes: Bacher (RFS, 2025)

The Gender Investment Gap over the Life Cycle

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1 One-sentence summary

This note summarizes Bacher’s paper, which uses SCF, PSID, and SCE data together with a structural life-cycle portfolio-choice model to show that single women invest less in risky assets than single men not because the model needs gender-specific preferences, but mainly because lower deterministic income, larger household sizes, and different future economic states change both the composition of households and the optimal policy functions governing risky investment.

2 Big picture

- **Question.** Why do single women hold lower equity shares and participate less in risky asset markets than single men over the life cycle?
- **Empirical fact.** In U.S. household data, single women are less likely to participate in the stock market and, especially when young, allocate a smaller fraction of net worth to risky assets.
- **Central explanation.** The model can match the gender investment gap without assuming that women are more risk averse. Instead, lower deterministic income and larger household sizes shift single women toward poorer states and make future consumption more vulnerable to financial shocks.
- **Two channels.** The paper separates a *composition effect* from a *policy effect*. The composition effect comes from single women being located in different states, such as lower wealth or larger households. The policy effect comes from different optimal choices conditional on the same current state, because future income, household size, marriage, divorce, medical expenses, and survival risks differ by gender.
- **Main quantitative result.** Deterministic income and household size are the most important determinants of the investment gap. Stochastic income differences work in the opposite direction: assigning women men’s more volatile income process would reduce women’s willingness to take financial risk and widen the investment gap.
- **Main methodological point.** Reduced-form regressions that control for current observable characteristics cannot fully account for the gap because forward-looking households condition on future income and household-size paths that are not directly controlled for in the regressions.
- **Economic takeaway.** Lower risky investment by women can arise from economic constraints and life-cycle background risk, rather than from intrinsic gender differences in preferences. This changes how one should interpret and evaluate policies aimed at female financial security.

3 Incremental contribution of the paper

3.1 From documenting gender gaps to explaining them structurally

Earlier work documents that women often hold safer portfolios, trade less, earn lower housing returns, or allocate less to equities in retirement accounts. This paper moves from documentation to mechanism. It asks whether a standard portfolio-choice model with common preferences can reproduce the gender investment gap once gender differences in income, household structure, and demographic transitions are taken seriously.

3.2 A life-cycle decomposition of the gender investment gap

The paper’s distinctive contribution is the decomposition of the gap into composition and policy effects. A reduced-form regression can control for current income, wealth, household size, education, and age, but it cannot easily control for the full future path of income, household size, marriage risk, divorce risk, retirement medical expenditures, and survival probabilities. The model makes those future states explicit and therefore shows why the unexplained part of the reduced-form gender coefficient need not be interpreted as preference heterogeneity.

3.3 A bridge between household finance and gender economics

The paper connects household portfolio choice to broader gender-related constraints. If women anticipate lower future earnings or larger future household sizes, the present value and risk profile of their human capital differ from men’s. Because human capital has bond-like features, this changes the optimal equity share. In this sense, the paper translates gender gaps in labor-market trajectories and household responsibilities into gender gaps in financial risk-taking.

3.4 Policy relevance

The findings imply that policies reducing the gender wage gap, subsidizing childcare, improving child support enforcement, or reducing female household background risk may affect wealth accumulation not only directly, but also indirectly by changing women’s optimal risky investment. The paper therefore cautions against interpreting lower female risky investment as purely preference-driven.

4 Data and measurement

4.1 Survey of Consumer Finances

The paper uses the 1989-2016 waves of the Survey of Consumer Finances (SCF) to measure household portfolio choices. The SCF is well suited for this part of the analysis because it contains detailed information on wealth, financial assets, risky assets, and household portfolio composition.

4.2 Panel Study of Income Dynamics

The paper uses the 1989-2017 waves of the Panel Study of Income Dynamics (PSID) to estimate income profiles, demographic processes, household sizes, marital transitions, and income shocks. The PSID is needed because the SCF is a repeated cross-section and does not provide the same panel structure for estimating individual income dynamics.

4.3 Survey of Consumer Expectations

The paper also uses the New York Fed Survey of Consumer Expectations (SCE), especially the 2014-2019 waves, to provide direct evidence on expected future earnings and expected household sizes. These data help validate the model mechanisms: single women expect lower future earnings and larger future nonspousal household sizes than single men.

4.4 Sample definitions

- A **single woman** is a family unit with a female head and no spouse present.
- A **single man** is analogously defined as a male head with no spouse present.
- **Couples** include legally married and cohabiting households.
- The main working-age sample contains singles aged 30-65 and couples whose average age is in the same range.
- Financial variables are converted into 2007 dollars and winsorized at the 1st and 99th percentiles.

4.5 Portfolio variables

The paper defines net worth as total assets minus total debt. Risky assets include direct stock holdings, corporate and foreign bonds, the fraction of mutual funds that include those assets, and the stock-invested fraction of retirement accounts. The main portfolio variables are:

- **Equity share:** risky assets divided by net worth. This combines participation and conditional risky allocation.
- **Participation rate:** whether the household owns any risky assets.
- **Conditional risky share:** risky assets as a fraction of net worth conditional on stock market participation.
- **Net worth:** total assets minus total debt.

5 Empirical facts

5.1 Life-cycle portfolio patterns

The paper first documents that single women hold lower equity shares than single men. The gap is largest early in the life cycle and narrows toward retirement. The lower equity share is driven primarily by lower participation rates, rather than by much lower risky shares conditional on participating. Couples have higher equity shares than singles because couples participate at higher rates.

5.2 Regression evidence

The paper estimates Tobit regressions of equity shares on a single-woman indicator, age polynomials, gender-age interactions, and controls such as education, household size, nonasset income, safe assets, year effects, and industry/occupation effects. The single-woman coefficient remains negative and statistically significant across specifications, especially for young households. The gap declines with age, which motivates a life-cycle explanation.

5.3 Why reduced-form controls are not enough

Even after controlling for current income, wealth, education, and household size, the gender coefficient remains negative. The structural model clarifies why: two people with the same current state may have different future income paths, future household sizes, marriage prospects, divorce risks, and retirement risks. These future states can change today's optimal portfolio choice.

6 Life-cycle model

6.1 Household types and timing

The model contains single women, single men, and couples. Life begins at age 30, retirement occurs at age 65, and the maximum age is 85. During working age, households face income risk and make consumption, saving, and portfolio-allocation decisions. During retirement, households receive pension income, face medical expenditures and survival risk, and value bequests.

At the beginning of each period, households observe income, marital status, asset returns, and cash-on-hand. They then choose consumption, safe-asset saving, and risky-asset saving. Investing in the risky asset requires paying a fixed participation cost each period.

6.2 Preferences

All households have CRRA preferences with common preference parameters across gender and marital status. For singles and couples,

$$u(c) = \eta_j \frac{(c/\eta_j)^{1-\gamma}}{1-\gamma},$$

where γ is relative risk aversion and η_j is an equivalence scale that adjusts for household size. Larger household size raises consumption needs and lowers effective patience in the model.

Households also value bequests:

$$\phi(a') = L \frac{(\omega + a')^{1-\gamma}}{1-\gamma},$$

where L governs the strength of the bequest motive and ω governs its luxury-good nature.

6.3 Income process

Income depends on gender, age, education, marital status, and stochastic income shocks:

$$y(i, j, \theta, \tilde{y}(\Omega)) = \bar{y}_i \theta_i \xi_{ij} \tilde{y}_{ij}(\Omega).$$

The stochastic component has persistent and transitory parts:

$$\tilde{y}_{ij} = z_{ij} + \epsilon_{ij},$$

with

$$z_{ij} = \begin{cases} \mu_z^{boom} + \rho_i z_{i,j-1} + \nu_{ij}, & \Omega = b, \\ \mu_z^{rec} + \rho_i z_{i,j-1} + \nu_{ij}, & \Omega = r. \end{cases}$$

The income process is estimated separately by gender and marital status. The key empirical pattern is that single women have lower deterministic income but a less volatile stochastic income process than single men.

6.4 Assets and returns

Households can invest in a safe asset and a risky asset. The safe return is constant. The risky return depends on the aggregate state. In booms it is drawn from a distribution with higher mean; in recessions it is drawn from a distribution with lower mean. The calibration uses an expected equity premium of 5 percent.

6.5 Marriage, divorce, and retirement risks

Single households face exogenous marriage probabilities that depend on gender, age, and education. Couples face divorce probabilities that depend on age and education. Upon divorce, assets are split equally in the benchmark, with 10 percent of assets destroyed. Retired households face gender- and age-specific medical expenses and survival probabilities. If one spouse dies, the survivor continues as a single with a share of the couple's assets.

6.6 Stock market participation cost

The model includes a fixed flow cost of stock market participation. The cost must be paid in any period in which the household invests in risky assets. It is allowed to rise with age after age 45. This cost helps the model match realistic stock market participation rates.

7 Estimation and calibration

7.1 First-stage estimation

The paper estimates or calibrates several components directly from data or the literature:

- deterministic income profiles from PSID data;
- stochastic income processes using a minimum-distance estimator;
- marital transitions using logit models;
- partner matching nonparametrically from the PSID;
- medical expenditures from prior HRS-based estimates;
- survival probabilities from U.S. Social Security Administration life tables;
- returns and recession probabilities from prior portfolio-choice literature and S&P 500 data;
- equivalence scales from observed household sizes using OECD weights.

7.2 Second-stage calibration

The remaining parameters are calibrated using the Simulated Method of Moments. The internally calibrated parameters are the discount factor β , relative risk aversion γ , and the young-household stock market participation cost S_{young}^F . The targeted moments are life-cycle profiles of equity shares, participation rates, and conditional risky shares for single men and single women.

The baseline calibration with the modified identity weighting matrix produces:

- annual discount factor $\beta = 0.917$;
- risk aversion $\gamma = 5.07$;
- participation cost of about \$288 for young households, rising to \$588 by age 65.

8 Quantitative results

8.1 Model fit

The model matches the life-cycle profiles of equity shares, participation rates, and conditional risky shares for both single women and single men. It also matches the net-worth profiles reasonably well through age 50, though it undershoots asset accumulation for older single men.

8.2 Simulated regressions

The paper repeats the reduced-form regressions on model-simulated data. Even though the model assumes common preference parameters across gender, the simulated regressions still show a negative single-woman effect that is strongest for young households and declines with age. This is central evidence that the unexplained reduced-form gender coefficient can arise from life-cycle constraints rather than preference heterogeneity.

8.3 Counterfactual decomposition

The paper replaces one female model element at a time with the corresponding male value. The most important mechanisms for the equity-share gap are deterministic income, household size, and initial wealth. Deterministic income explains about half of the equity-share gap. Household size explains more than the full gap and can reverse it. Initial wealth also explains a substantial part. By contrast, assigning women men's stochastic income process widens the gap because men's income risk is higher.

8.4 Composition and policy effects

The composition effect explains most of the gender gap in participation, especially among young households. Women are more likely to be in lower-wealth or larger-household states, where participation is less likely. The policy effect is more important for the conditional risky share among young households. Conditional on the same current state, future deterministic income and future household size affect the optimal equity share.

9 Economic intuition

9.1 Human capital as a bond-like asset

In life-cycle portfolio theory, labor income often acts like a bond-like asset because it is less risky than equity and only mildly correlated with stock returns. A person with more safe future labor income can optimally hold more equity in financial wealth. Since single women have lower expected future deterministic income, they have less human capital and optimally take less financial risk, even conditional on current wealth.

9.2 Household size and effective patience

Larger household size raises current and future consumption needs. In the model, this appears through the equivalence scale in utility and lowers the effective discount factor. Single women, especially younger single women, are more likely to have larger household sizes because of children or other nonspousal household members. This makes them more vulnerable to financial shocks and reduces optimal risky investment.

9.3 Why stochastic income risk works differently

One might expect women to invest less because they face more income risk. The model finds the opposite: single women have a less volatile income process than single men. If women were assigned men's more volatile stochastic income process, they would become less willing to take financial risk, widening the gender investment gap. Therefore, stochastic income differences are not the main explanation for lower female risky investment; they partly offset the gap.

10 Identification and interpretation

10.1 What is identified by the reduced-form facts

The reduced-form regressions identify robust conditional gender differences in portfolio allocations. They show that observable current characteristics do not fully account for the gap. However, the regressions cannot identify whether the remaining gap reflects preferences, expectations, constraints, or future risks.

10.2 What the structural model adds

The structural model imposes a disciplined life-cycle framework and asks whether common preferences can rationalize the observed patterns. The answer is yes. The model decomposes the gap into mechanisms and shows that current and future economic constraints can generate the observed gender investment gap.

10.3 Limits of the interpretation

The model does not prove that there are no gender differences in preferences. Rather, it shows that such differences are not necessary to rationalize the observed investment gap. The paper's interpretation is therefore constructive: a rich but standard economic environment can produce the gap without changing risk aversion by gender.

11 Tables and figures

11.1 Figure 1: Life-cycle patterns of household finances (data) and Table 1: Regression coefficients and marginal effects: Equity shares of singles

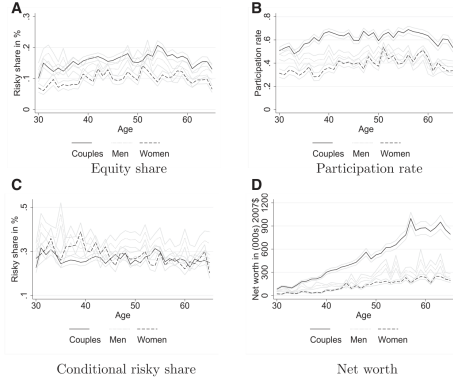


Figure 1

(a) **Figure 1.** Life-cycle profiles of equity shares, participation rates, conditional risky shares, and net worth in the SCF.

Table 1
Regression coefficients and marginal effects: Equity shares of singles

	(1) Equity share	(2) Equity share	(3) Equity share	(4) Equity share
Single woman	-0.3359*** (0.0131)	-0.2151*** (0.0113)	-0.1671*** (0.0123)	-0.1239*** (0.0140)
Single woman*age	0.0052*** (0.0002)	0.0032*** (0.0002)	0.0026*** (0.0002)	0.0015*** (0.0002)
Age	-0.0995*** (0.0135)	-0.0508*** (0.0146)	-0.0483*** (0.0165)	-0.0283 (0.0185)
age ² * 100	0.2457*** (0.0292)	0.1357*** (0.0316)	0.1188*** (0.0356)	0.0739* (0.0408)
age ³ * 10000	-0.1947*** (0.0206)	-0.1123*** (0.0222)	-0.0968*** (0.0249)	-0.0592** (0.0291)
High education		0.2698*** (0.0020)	0.2469*** (0.0019)	0.1309*** (0.0042)
Number of HH members		-0.0556*** (0.0018)	-0.0558*** (0.0020)	-0.0509*** (0.0030)
Nonasset income		0.0324*** (0.0007)	0.0288*** (0.0006)	0.0257*** (0.0007)
Safe assets			0.0218*** (0.0012)	0.0146*** (0.0009)
Constant	1.2163*** (0.2027)	-0.0393 (0.2168)	-0.1261 (0.2398)	-0.4092 (0.2749)
Observations	10,244	10,240	10,240	7,607
Year FE	No	Yes	Yes	Yes
Industry and occupation FE	No	No	No	Yes
ME for women at age 30	-0.1814*** (0.0065)	-0.1179*** (0.0062)	-0.0892*** (0.0064)	-0.0790*** (0.0075)
ME for women at mean age (47)	-0.0919*** (0.0032)	-0.0616*** (0.0036)	-0.0441*** (0.0035)	-0.0551*** (0.0042)
ME for women at age 65	-0.0011 (0.0031)	-0.0045 (0.0028)	0.0016 (0.0032)	-0.0267*** (0.0018)

This table reports regression results from Tobit estimations on the sample of individuals who live in households with no spouse present. Dependent variable is the unconditional risky share (equity share). The equity share is defined as the fraction of net worth that is invested in risky assets. *Single woman* is a dummy indicating that the household head is a woman. *High education* is a dummy equal to one if the household head has more than 12 years of education. *Safe assets* refers to all net worth that is not invested in risky assets. "ME" indicates the marginal effect of being a woman at the respective age. Data come from the waves 1989 until 2016 of the Survey of Consumer Finances (SCF). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(b) **Table 1.** Reduced-form Tobit regressions of equity shares for singles.

Interpretation of Figure 1. Figure 1 documents the baseline empirical pattern. Single women hold lower equity shares than single men, and the difference is especially visible at young ages. The participation-rate panel shows that the gender gap is driven mostly by the extensive margin: single women are less likely to hold risky assets at all. The conditional risky-share panel shows a smaller and noisier gender difference. The net-worth panel shows that single women accumulate less wealth, which is important because lower wealth also mechanically reduces the likelihood of paying fixed participation costs.

Interpretation of Table 1. Table 1 shows that the gender investment gap survives many current-observable controls. The coefficient on *Single woman* is negative across all specifications, and the interaction with age is positive, meaning the gap shrinks with age. Adding education, household size, nonasset income, safe assets, and industry/occupation fixed effects reduces but does not eliminate the gap. This motivates the structural model: current observables alone do not fully explain the gap.

11.2 Figure 2: Life-cycle profiles of the deterministic income component and Table 2: Estimation results: Stochastic income component

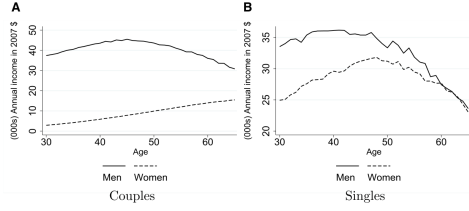


Figure 2
Life cycle profiles of the deterministic income component
This figure plots the life-cycle profiles for the deterministic income component by gender and marital status. Data come from the waves 1989 until 2016 of the Panel Study of Income Dynamics (PSID).

I follow [Borella, De Nardi, and Yang \(2020\)](#) and first split the sample by marital status and then separately regress the inverse hyperbolic sine of income

Table 2
Estimation results: Stochastic income component

Parameter	Singles		Couples	
	Men	Women	Men	Women
ρ	0.9522 (0.0079)	0.9348 (0.0095)	0.9353 (0.0046)	0.9273 (0.0045)
σ_ϵ^2	0.0899 (0.0115)	0.0856 (0.0118)	0.0809 (0.0051)	0.1583 (0.0092)
σ_η^2	0.1689 (0.0341)	0.1547 (0.0217)	0.1412 (0.0109)	0.2851 (0.0177)

This table reports the estimation results for the stochastic part of the income process. Standard errors in parentheses.

(a) **Figure 2.** Deterministic income profiles by gender and marital status.

(b) **Table 2.** Estimated stochastic income process by gender and marital status.

Interpretation of Figure 2. Figure 2 shows the deterministic income profiles that become central to the model. For singles, men have higher deterministic income than women for most of working life, and the difference is largest early in life. This timing matters because early participation in risky assets generates higher expected returns and wealth accumulation over subsequent years.

Interpretation of Table 2. Table 2 reports the estimated stochastic income processes. Single women have slightly lower persistence and lower persistent and transitory shock variances than single men. This means women's stochastic income risk is not the reason they invest less. In fact, replacing female stochastic income with male stochastic income makes women more exposed to background risk and tends to widen the equity-share gap.

11.3 Table 3: Second-stage parameters and Figure 3: Model fit of investment patterns (singles)

4.2 Model fit

4.2.1 Life-cycle profiles of household finances Figure 3 shows that the model matches well the life-cycle profiles of equity shares, participation rates, and conditional risky shares for both single men and single women.

Table 3
Second-stage parameters

W	β	γ	S_{young}^E	S_{old}^E
$\mathcal{I}$	0.917	5.07	\$288	\$588
Ψ	0.918	5.31	\$228	\$528

This table lists the values for internally calibrated model parameters. W denotes the weighting matrix, as explained in Section 3.2.

(a) **Table 3.** Calibrated structural parameters.

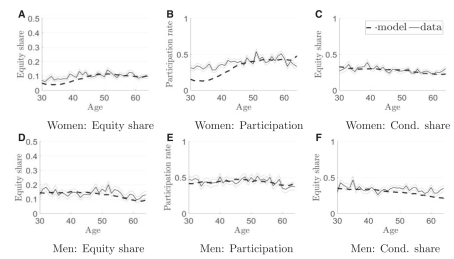


Figure 3
Model fit of investment patterns (singles)
This figure plots the model fit of equity shares, participation rates, and conditional risky shares for single women and single men. The solid lines represent the data (including 95% confidence bands, as plotted in Figure 1) whereas the dashed lines display the simulated life-cycle profiles generated from the model.

(b) **Figure 3.** Model fit for equity shares, participation, and conditional risky shares.

Interpretation of Table 3. Table 3 reports the second-stage parameters. The estimates imply moderate/high risk aversion and a realistic participation cost. The participation cost is not gender-specific; it is common across household types. Therefore, the model's ability to match gender differences cannot come from assuming women face a higher stock market participation cost.

Interpretation of Figure 3. Figure 3 compares model-generated and data life-cycle profiles for single women and single men. The model fits equity shares, participation rates, and conditional risky shares reasonably well for both groups. This is the key quantitative validation: without gender-specific preferences, the model can replicate the broad empirical gender investment gap.

11.4 Figure 4: Model fit of net worth (singles) and Table 4: Regression coefficients and marginal effects: Data versus model simulations

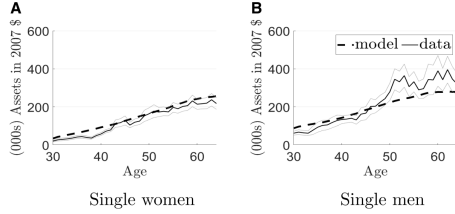


Figure 4
Model fit of net worth (singles)
This figure plots the model fit of asset accumulation (expressed as net worth) for single women and single men. The solid lines represent the data (including 95% confidence bands, as plotted in Figure 1) whereas the dashed lines display the simulated life-cycle profiles generated from the model. Data lines are smoothed using a three-point moving average.

Table 4
Regression coefficients and marginal effects: Data versus model simulations

	(1) Equity share		(3) Participation rate		(5) Conditional share	
	Model	Data	Model	Data	Model	Data
Single woman	-0.5337*** (0.0346)	-0.1671*** (0.0123)	-0.4328*** (0.0366)	-0.1363*** (0.0116)	-0.0771*** (0.0132)	-0.0195 (0.0170)
Single woman*age	0.0092*** (0.0007)	0.0026*** (0.0002)	0.0074*** (0.0008)	0.0026*** (0.0002)	0.0012*** (0.0003)	-0.0008** (0.0004)
Age	0.1654*** (0.0273)	-0.0483*** (0.0165)	-0.0072 (0.0296)	-0.0917*** (0.0147)	0.0663*** (0.0097)	0.0307*** (0.0118)
age ² * 100	-0.3784*** (0.0586)	0.1188*** (0.0356)	-0.0319 (0.0641)	0.2123*** (0.0324)	-0.1517*** (0.0206)	-0.0591** (0.0242)
age ³ * 10000	0.2741*** (0.0409)	-0.0968*** (0.0249)	0.0460 (0.0450)	-0.1611*** (0.0231)	0.1099*** (0.0143)	0.0394** (0.0163)
High education	0.0374*** (0.0074)	0.2469*** (0.0019)	0.0235*** (0.0082)	0.2585*** (0.0018)	0.0288*** (0.0027)	0.0547*** (0.0032)
No. of HH members		-0.0558*** (0.0020)		-0.0407*** (0.0008)		-0.0234*** (0.0012)
Nonasset income	0.0418*** (0.0036)	0.0288*** (0.0006)	0.0390*** (0.0040)	0.0261*** (0.0004)	0.0060*** (0.0012)	0.0038*** (0.0006)
Safe assets	0.1226*** (0.0039)	0.0218*** (0.0012)	0.1848*** (0.0039)	0.0147*** (0.0003)	-0.0574*** (0.0013)	-0.0380*** (0.0006)
Constant	-4.1235*** (0.4178)	-0.1261 (0.2398)	-1.6514*** (0.4446)	1.1751*** (0.2160)	0.0384 (0.1482)	0.1063 (0.1882)
Observations	10,240	10,240	10,240	10,240	3,900	4,285
Year FE	No	Yes	No	Yes	No	Yes
ME at age 30	-0.2585*** (0.0148)	-0.0892*** (0.0064)	-0.2110*** (0.0152)	-0.0590*** (0.0048)	-0.0418*** (0.0058)	-0.0440*** (0.0064)
ME at mean age (47)	-0.1026*** (0.0074)	-0.0441*** (0.0035)	-0.0853*** (0.0081)	-0.0143*** (0.0018)	-0.0217*** (0.0027)	-0.0593*** (0.0026)
ME at age 65	0.0626*** (0.0138)	0.0016 (0.0032)	0.0477*** (0.0159)	0.0311*** (0.0040)	-0.0005 (0.0048)	-0.0727*** (0.0072)

This table reports regression results from estimations on the sample of individuals who live in households with no spouse present. Columns "Model" are model simulations, whereas columns "Data" refer to data from the SCF waves 1989 until 2016. Equity share = Unconditional risky share. Participation rate = Stock market participation rate. Conditional share = Conditional risky share. *Single woman* is a dummy indicating that the household head is a woman. *High education* is a dummy equal to one if the household head has more than 12 years of education. *Safe assets* refers to all net worth that is not invested in risky assets. "ME" indicates the marginal effect of being a woman at the respective age. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(a) **Figure 4.** Model fit for net worth of single women and men.

(b) **Table 4.** Reduced-form regressions in model-simulated data versus SCF data.

Interpretation of Figure 4. Figure 4 shows the model fit for asset accumulation. The model matches single women's net worth closely and matches single men's net worth reasonably well until about age 50, but underpredicts older single men's wealth. This matters for interpreting the wealth-gap decomposition, though the paper notes that the relative importance of the main mechanisms is similar when focusing on ages below 50.

Interpretation of Table 4. Table 4 is one of the paper's most important diagnostic exercises. The model assumes common preferences, yet regressions on simulated data still produce a negative single-woman coefficient and a shrinking gender gap with age. This means that a reduced-form gender coefficient should not automatically be read as evidence of gender-specific risk aversion. Life-cycle state variables and future risks can generate the same empirical pattern.

11.5 Table 5: Decomposition results and Figure 5: Composition versus policy effect: Aggregate

Gap in asset holdings in (000s) 2007 \$	Model	% explained	Data
Baseline	48.31		91.00 ^a
Male deterministic income	24.64	48.99	
only: constant (γ)	-106.71	320.89	
age component (ζ)	95.47	-97.62	
education premium (θ)	57.95	-19.96	
Male deterministic income: slope effect	43.31	10.35	
Male deterministic income: level effect	30.05	37.81	
Male stochastic income	22.44	53.55	
Male household size	27.14	43.83	
Male marriage probability	43.51	9.93	
Male marriage market	45.02	6.82	
Male education distribution	44.55	7.78	
Male medical expenses	48.54	-0.47	
Male survival probability	49.04	-1.51	
Male initial wealth	33.11	31.47	
Gap in equity share in % - points	Model	% explained	Data
Baseline	4.31		4.24
Male deterministic income	2.09	51.39	
only: constant (γ)	-6.34	247.12	
age component (ζ)	8.45	-96.32	
education premium (θ)	4.97	-15.50	
Male deterministic income: slope effect	3.81	11.64	
Male deterministic income: level effect	2.57	40.30	
Male stochastic income	5.72	-32.87	
Male household size	-1.17	127.21	
Male marriage probability	3.52	18.30	
Male marriage market	3.02	29.80	
Male education distribution	4.15	3.68	
Male medical expenses	4.25	1.21	
Male survival probability	4.50	-4.38	
Male initial wealth	1.88	56.33	

This table shows the results of the decomposition exercise. The column "Model" reports the average gender gap in the respective counterfactual. The column "% explained" indicates how much of the baseline gap can be explained through that channel. All values refer to averages over the life cycle.

^aThe lower wealth gap in the model compared to the data mainly arises from the model's inability to explain the fast asset accumulation of single men beyond age 50 (see Figure 4). When replicating the decomposition exercise on the sample of households below 50 (i.e., for which the model matches the data better), the relative importance of the respective channels hardly changes.

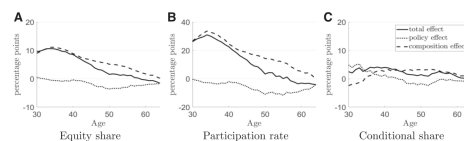


Figure 5
Composition versus policy effect: Aggregate

This figure decomposes aggregate gender gaps in (unconditional) equity shares, participation rates, and conditional risky shares into a composition and policy effect. The composition effect (dashed line) shows what part of the overall gap can be explained through gender differences in the sample composition whereas the policy effect (dotted line) shows what part of the gap can be explained by differences in policy functions, conditional on state variables.

(a) **Table 5.** Counterfactual decomposition of asset-holding and equity-share gaps.

(b) **Figure 5.** Composition versus policy effects for aggregate equity shares, participation, and conditional shares.

Interpretation of Table 5. Table 5 decomposes the model-implied gender gaps. Deterministic income explains about 51 percent of the equity-share gap and about 49 percent of the wealth gap. Household size explains more than the full equity-share gap and 44 percent of the wealth gap. Initial wealth explains 56 percent of the equity-share gap. Stochastic income has a negative explained share for the equity gap, meaning it works against the gap: men's more volatile stochastic income would make women invest even less if assigned to them. The table therefore identifies deterministic income, household size, and initial wealth as the main forces.

Interpretation of Figure 5. Figure 5 separates aggregate gender differences into composition and policy effects. Composition explains much of the participation gap, especially early in life: single women are more often in states where participation is less likely. For the conditional risky share, however, the policy effect is more important among young households, showing that even conditional on current states, forward-looking policy functions differ because future outcomes differ.

11.6 Figure 6: Further decomposition of policy effect and Table 6: Expected earnings of single households: SCE data

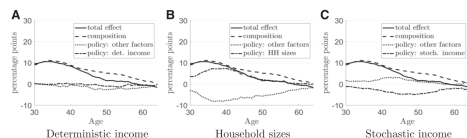


Figure 6
Further decomposition of policy effect
This figure illustrates the importance of gender heterogeneity in deterministic income, household sizes, and stochastic income for the policy effect in unconditional equity shares. The solid lines refer to the total gender investment gap, the dashed line to the composition effect, the dashed-dotted lines represent the part of the policy effect that can be attributed to the respective channel, whereas the dotted lines report the importance of all remaining channels (as listed in Table 5).

Table 6
Expected earnings of single households: SCE data

	(1)	(2)	(3)	(4)
	log(exp. earnings)	log(exp. earnings)	log(exp. earnings)	log(exp. earnings)
			Age ≤ 45	Age > 45
Single woman	-0.1619*** (0.0376)	-0.0922** (0.0464)	-0.2394*** (0.0548)	-0.0915** (0.0445)
log(current earnings)	0.6556*** (0.0502)	0.5792*** (0.0670)	0.5712*** (0.0648)	0.7927*** (0.0550)
High education	0.2617*** (0.0365)	0.2132*** (0.0481)	0.3591*** (0.0524)	0.1429*** (0.0382)
1. age > 40	-0.0207 (0.0421)	-0.0186 (0.0460)		
Financial wealth		0.0246** (0.0095)		
Constant	3.5953*** (0.5231)	4.2170*** (0.6941)	4.4481*** (0.6874)	2.1708*** (0.5859)
Observations	3,009	1,697	1,774	1,235
Year FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes

This table reports regression results from OLS estimations on the sample of individuals who live in households with no spouse present. Dependent variable is the log of expected earnings in four months. *Single woman* is a dummy indicating that the household head is a woman. *High education* is a dummy equal to one if the household head has more than 12 years of schooling. Data come from the Labor Market Module of the Survey of Consumer Expectations (SCE), waves 2014 until 2019. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(a) **Figure 6.** Further decomposition of the policy effect.

(b) **Table 6.** Expected earnings of single households in SCE data.

Interpretation of Figure 6. Figure 6 decomposes the policy effect into deterministic income, household size, and stochastic income. Lower future deterministic income leads women to choose lower risky shares conditional on the same current state. Larger household size also pushes women toward lower risky investment because future consumption needs are higher. Stochastic income works in the opposite direction: because women's stochastic income is less risky, it tends to increase their willingness to hold equity relative to a counterfactual with men's stochastic risk.

Interpretation of Table 6. Table 6 provides direct survey evidence consistent with the model. Conditional on current earnings, education, age, year, and region, single women expect lower future earnings than single men. The gap is larger among younger singles, where the model says deterministic income differences are most important. This supports the interpretation that future income expectations are part of the investment gap.

11.7 Figure 7: Expected household sizes of singles: SCE data and Table 7: Robustness checks

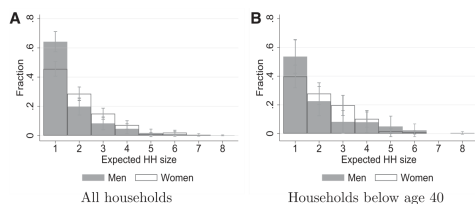


Figure 7
Expected household sizes of singles: SCE data
This figure plots the histogram of expected household sizes (= expected number of household members) for single households. Panel A plots the distribution for the entire sample, whereas panel B restricts the sample to households below age 40. Data are from the Household Finance Module of the Survey of Consumer Expectations (SCE), waves 2014 until 2019.

Table 7
Robustness checks

Gender gap in ...	Asset holdings (in 000s \$)	Equity shares (in % points)
Baseline	48.31	4.31
<i>A. Alimony and child support</i>		
Modified splitting rule after divorce	28.55	2.65
<i>B. Bequests</i>		
Variation in bequest parameters	48.42	4.29
Lump-sum transfer at age 55	48.41	4.17
<i>C. Exogenous parameters</i>		
Higher correlation of income shocks within couples	48.50	3.93
No asset drop after death of spouse	48.10	4.22
Reduced medical expenses for couples	47.90	4.18
Higher pension payments for couples	47.79	4.05

This table reports the average gender gap in asset holdings and equity shares in the baseline model, as well as in alternative versions: panel A performs robustness with regard to alimony payments and child support, panel B with regard to bequests, and panel C with regard to exogenously set model parameters. "Gender Gap" describes the difference of the respective variable between single men and single women, averaged over working age. The gender gap in asset holdings is expressed in (000s) 2007s. The gender gap in equity shares is expressed in percentage points.

(a) **Figure 7.** Expected household sizes of singles in SCE data.

(b) **Table 7.** Robustness checks for alimony, bequests, and exogenous model parameters.

Interpretation of Figure 7. Figure 7 shows that single women are less likely than single men to expect to live alone and more likely to expect additional household members. This is especially clear among younger households. The evidence supports the model’s assumption that single women face larger expected household sizes, which affects consumption needs and lowers optimal risky investment.

Interpretation of Table 7. Table 7 shows that the main gender investment gap is robust to alternative assumptions. Changing the post-divorce asset split in favor of women narrows the gap but does not eliminate the mechanism. Varying bequest assumptions and exogenous parameters such as within-couple income-shock correlation, surviving-spouse asset retention, medical expenses, and pension payments leaves the equity-share gap close to the baseline. This increases confidence that the main results are not an artifact of one narrow calibration choice.

11.8 Table 8: Distribution of (expected) bequests by family type (data)

Table 8
Distribution of (expected) bequests by family type (data)

		Singles	
	Couples	Men	Women
<i>A. Bequests received</i>			
% received inheritance (> 55 yrs)	30.23 (0.36)	25.61 (0.73)	26.25 (0.55)
\$ amount received in (000s) (> 55 yrs)	56.16 (1.52)	35.70 (2.61)	30.43 (1.78)
% expect inheritance	17.05 (0.14)	14.87 (0.28)	10.24 (0.19)
<i>B. Bequests given</i>			
% perceive bequests as important	51.81 (0.18)	52.25 (0.41)	53.39 (0.31)
% expect to give bequest	56.06 (0.18)	51.61 (0.40)	41.75 (0.31)

This table reports the distribution of (expected) bequests across family types. “% received inheritance (> 55 yrs)” indicates the fraction of HHs above age 55 that have received an inheritance or substantial gift, whereas “\$ amount received in (000s) (> 55 yrs)” indicates the average amount. “% expect inheritance” denotes the fraction of households that expect to receive an inheritance in the future. “% perceive bequests as important” is the fraction who considers leaving something behind as important, and “% expect to give bequest” is the fraction who expects to do so. Standard errors are in parentheses. Data come from the Survey of Consumer Finances (SCF), waves 1989-2016.

distribution of these (expected) inheritances by family type. Single men and

Table 8. Distribution of received, expected, and intended bequests by family type.

Interpretation of Table 8. Table 8 supports the paper’s treatment of bequests. Single men and single women report similar probabilities of having received inheritances, though men receive somewhat larger amounts conditional on receiving. The importance attached to leaving bequests is also similar across groups. This makes it less plausible that gender-specific bequest preferences are driving the main investment gap.

12 Short summary

- The paper studies why single women invest less in risky assets than single men.
- The empirical gender investment gap is strongest among young households and is driven mainly by lower stock market participation.
- Reduced-form regressions with controls reduce but do not eliminate the gap.
- The paper builds a life-cycle portfolio-choice model with singles, couples, marriage, divorce, income risk, retirement, medical expenses, survival risk, bequests, and stock market participation costs.
- Preferences are held constant across gender and marital status.
- The model matches life-cycle investment patterns for single men and single women.

- Counterfactual exercises show that lower female deterministic income and larger female household sizes are the main determinants of the gap.
- Deterministic income affects the gap through both its level and life-cycle slope.
- Household size matters because it raises consumption needs and lowers effective patience, reducing willingness to take financial risk.
- Stochastic income differences do not explain the gap; they partly offset it because single women have less volatile income.
- Composition effects explain much of the participation gap, while policy effects matter especially for young households' conditional risky shares.
- SCE evidence confirms that single women expect lower future earnings and larger household sizes than single men.

13 Conclusion

13.1 Core conclusion

Bacher shows that the gender investment gap can be rationalized within a standard life-cycle portfolio-choice model without introducing gender heterogeneity in preferences. Single women invest less in risky assets mainly because they face lower deterministic income, larger household sizes, lower initial wealth, and different future economic states. These differences place them in lower-wealth states and change their optimal portfolio choices conditional on current states.

13.2 Economic intuition

The central intuition is that optimal risky investment depends not only on current wealth and income, but also on future human capital and future consumption needs. Single women expect lower future deterministic income and larger household sizes. Lower expected future income reduces the bond-like component of their total wealth, and larger household sizes raise future consumption needs. Both mechanisms make single women more vulnerable to financial shocks and reduce optimal equity exposure.

13.3 Incremental contribution revisited

The paper contributes by showing that a gender coefficient in reduced-form investment regressions need not imply gender-specific risk preferences. It may instead reflect forward-looking responses to gendered life-cycle constraints. The paper therefore reframes the gender investment gap as an outcome of income trajectories, household structure, and dynamic background risk.

13.4 Implications

The findings imply that policies improving women's labor-market trajectories, reducing childcare or household-resource burdens, or strengthening income support may increase female risky investment and wealth accumulation. The effect of such policies may be amplified by higher expected portfolio returns if women respond by increasing stock market participation and equity shares.

13.5 Limitations

The paper does not rule out gender differences in preferences; it shows they are not required for the model to match the data. The model abstracts from some institutional details such as explicit alimony histories, endogenous labor supply, housing, financial advice, and learning. Some model moments, especially older single men's asset accumulation, are not perfectly matched. Nevertheless, the robustness exercises suggest that the main mechanisms are stable.

13.6 Takeaway

The main takeaway is that lower female risky investment can be generated by economic constraints and future household risks rather than innate risk aversion. The gender investment gap is therefore not only a household-finance puzzle; it is also a life-cycle gender-economics problem linking income, family structure, background risk, and wealth accumulation.